

Assignment 8: Special Repo Files

Overview

Welcome back from Spring break! For this assignment, you will be continuing to practice everything that you have learned in this course regarding Git and GitHub (you didn't forget everything over break, right?), and there is now the addition of Git Ignore files and Read Me files.

Assignment Instructions

Each student is to work on this assignment individually, however there is a part where you will need to ask a peer to make a commit to your repo.

Part 1: Repo Setup

Create a new repository for this assignment called **FizzBuzz**.

Create a file to start off the program, such as a **.java** file that contains a **main** method but nothing else – essentially a program that can be run, but does nothing. Make sure you actually run this “blank” program before moving on!

Create a Git Ignore file. Populate it with some files or folders that you know ahead of time should be ignored. If you are unsure of what to put, you can find the standard Git Ignore file I use for Java projects [here](#) and copy it (it is setup to ignore auto-generated files from Eclipse and IntelliJ IDEs, as well as some other common files that should be ignored). If you are using a programming language other than Java and are unsure of what to put, either Google for a pre-made Git Ignore file, or ask me for some pointers.

Create a Read Me file. Create a heading with the text FizzBuzz. Leave it at that for now.

Run a **git status** to ensure there are no files that you want to ignore popping up. If you see any, update your Git Ignore file, and then run another **git status** to confirm that it is being ignored. You should only have three files going into this initial commit – the “blank” code file (such as a **.java** file), the Git Ignore file, and the Read Me file. If you see anything else, it means you need to adjust your Git Ignore file.

Push all of this up to GitHub directly to the **master** branch as the initial commit(s). Then, update your repo's settings to disallow pushes directly to the **master** branch (aka pull requests are required).

Part 2: FizzBuzz It Up!

If you do not know, FizzBuzz is a famous coding interview question that eliminates a ton of candidates right off the bat, so it will be good for you to experience writing this. It's an extremely simple program, but you'd be surprised at how many computer science students make it through four years of school without being able to solve it.

Create a new branch named **solve-fizzbuzz**. Then, in the programming language of your choice, write a program that prints out every number between 1 and 100 (inclusive). If the number is divisible by 3, print out **Fizz** instead of the number. If the number is divisible by 5, print out **Buzz** instead of the number. If the number is divisible by both 3 and 5, print out **FizzBuzz** instead of the number.

Update the Read Me file to include a heading (lower level than the existing header from the previous step) that says **What is this project?**, and then type a very detailed explanation of what the program does as if a very beginner programmer was reading it. Be sure to use proper styling where appropriate to make it easy to comprehend.

Once the program meets requirements, create a pull request, and merge it.

Part 3: Actively Ignoring

Ask a peer for a quick hand for this part. Add them to your repo. They are to clone your repo, create a new branch called **password**, and create a new file called **ALL_OF_MY_PASSWORDS.txt**. Inside of it, they should add some random text (please do not put any actual passwords). Finally, they should then commit, push, create a pull request, and merge in the pull request.

Uh oh! There is now password data that is showing up in the repo. Obviously, these passwords are now compromised. Assuming you already have changed any password that was compromised (in this hypothetical scenario), you still need to remove the file **ALL_OF_MY_PASSWORDS.txt** from the repo.

Your task is to successfully remove this file from the repo, and ensure that it is added to the Git Ignore file so that it cannot be accidentally added to the repo again by you or anyone else.

Submission

Your final submission to Blackboard should include:

- A link to your GitHub repository's webpage

Grading Rubric

In order to get an A on this assignment (20/20):

- You followed directions for setting up the repo: **6pts**
- Your FizzBuzz program and Read Me file meets requirements: **7pts**
- You successfully removed the passwords file from the repo and set things up so that it cannot be included again by you or other developers: **7pts**

Note: I will additionally take one point off for every file that is in your repo that should have been excluded and added to the Git Ignore file.